

web animation and motion

THE DECK AND THE SPEAKER NOTES · 36 SLIDES

TALK

Key Clark

DATE

17 September 2026

MADE FOR

Folklore Digital

PAPER AND LIVE EXAMPLES

wam-2026.netlify.app

Every example in this deck is live running code, not a recording. The stills here are captured from those same demos.

- **Why I'm up here**

- Experienced engineer; enabling the *what* of design is where I add most value
- Knew straight away the talk had to be about **opportunity and friction**

- **The nine months**

- Projects, experiments and pit stops on the design ↔ engineering workflow around motion
- This talk is a consequence of those; it has taken a lot of work

- **Made for Folklore Digital**

- The paper, the talk and every example were made here, for the people I build with

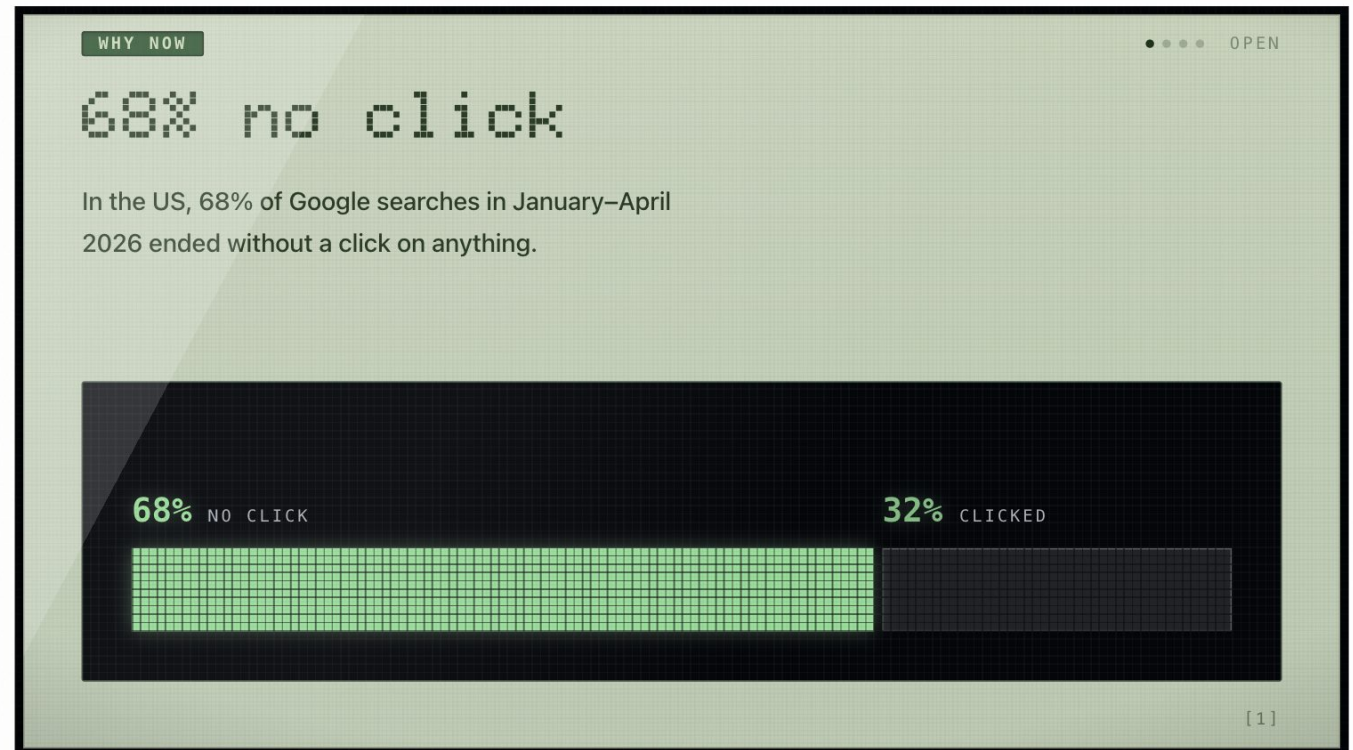
- **Everything today is live, running code**

- Every source is in the paper on the site



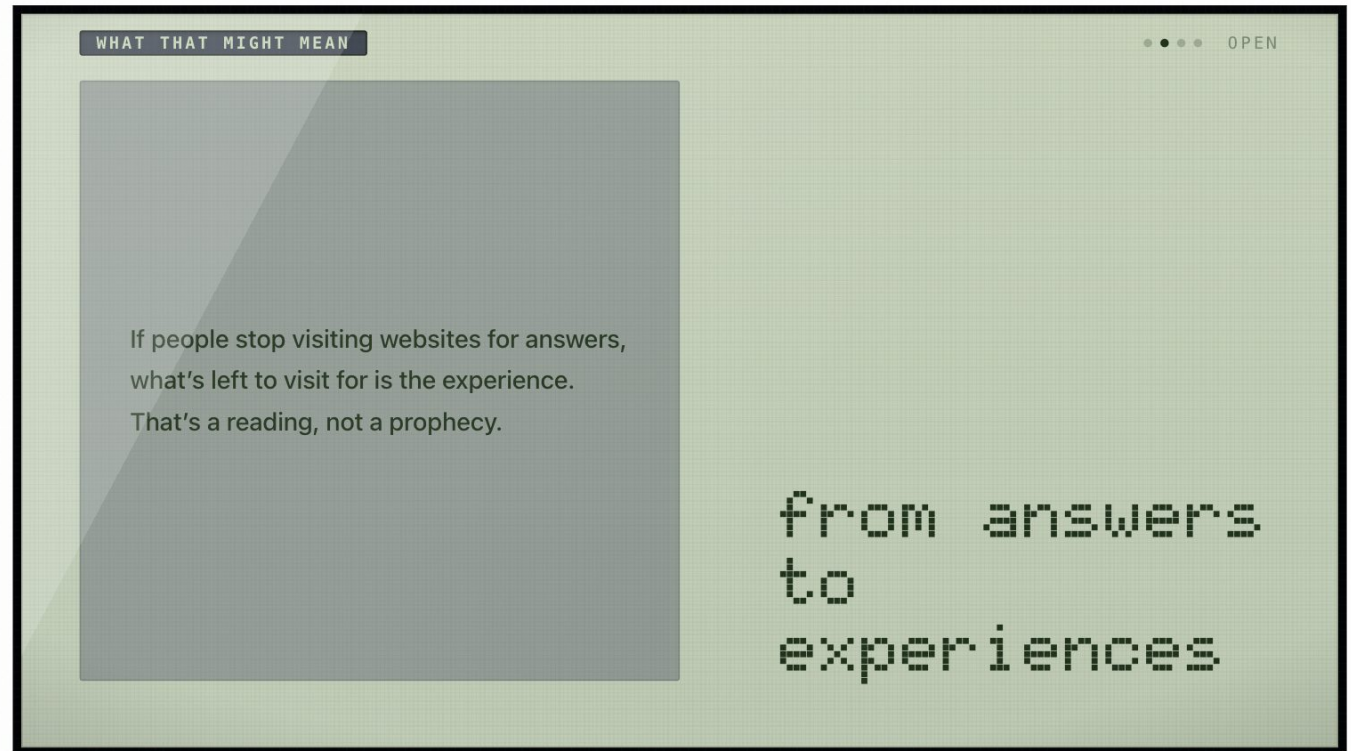
A tour through the ways a web page can move, and what it takes to get them built.

- **68% of US Google searches ended without a click**
 - January–April 2026 [1]
 - SparkToro's analysis of Similarweb's desktop + mobile-web panel
 - "No click" means no click on anything, ads and Google's own pages included
- **Searches producing any click fell 9.51 points, 2024 → 2026**
 - A 22.9% relative decline [1]
- **AI summaries mean fewer clicks**
 - Pew: a result clicked in **8%** of visits with a summary vs **15%** without [2]
 - Ahrefs: **58%** lower click-through for the top result [3]
 - Size varies by method; one tracker shows the gap narrowing in 2026 [4]
- **Google disputes the aggregate**
 - Says total clicks to websites are "relatively stable"; no dataset published [5]
 - Studies measure *clicks per search*; Google describes *total volume*
- **A reading, not a prophecy**



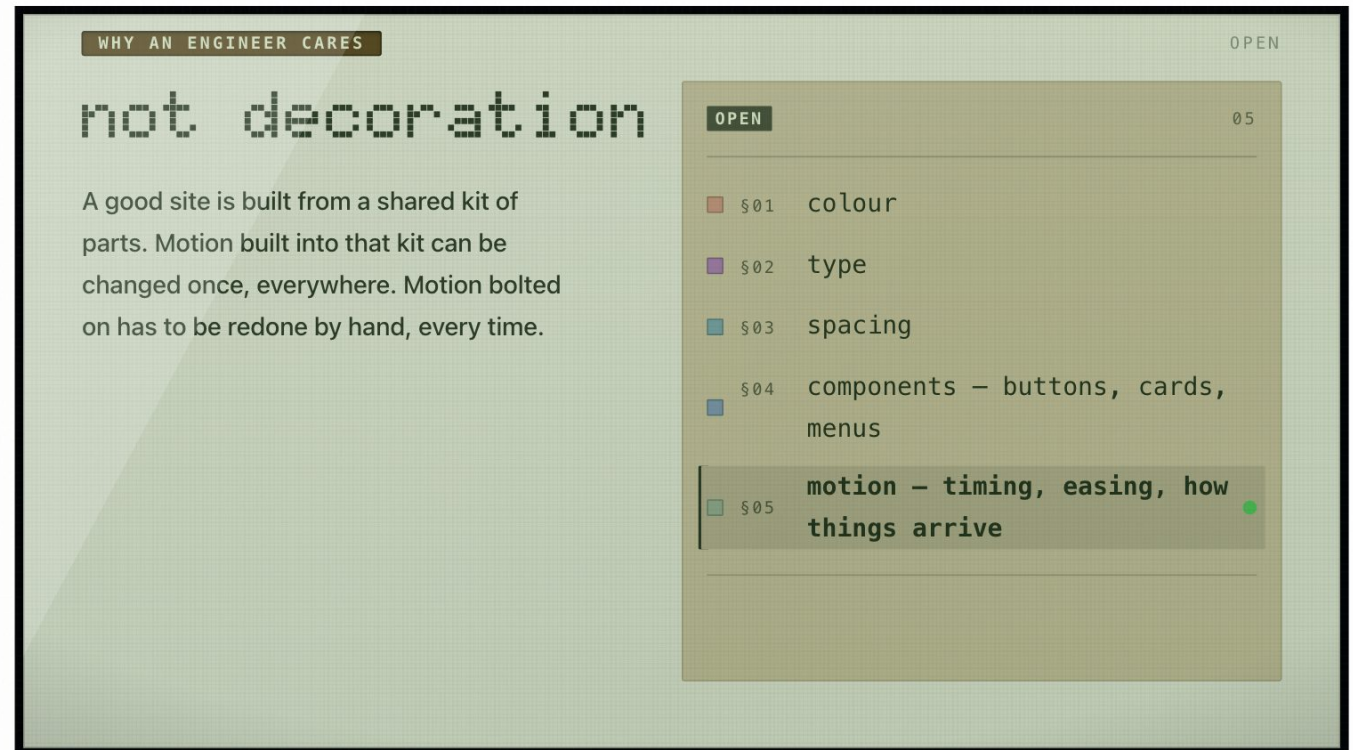
In the US, 68% of Google searches in January–April 2026 ended without a click on anything.

- **If the web stops being where people go for answers, it becomes where they go for experiences**
 - An observation on the table, not a fact
- **What a chat can't deliver: authored sequence**
 - Someone decided what you see
 - In what order
 - How it feels arriving
- **A different product, not a nicer version of the same one**
 - The techniques are how you build it
- **Do people want these experiences?**
 - Deliberately left open



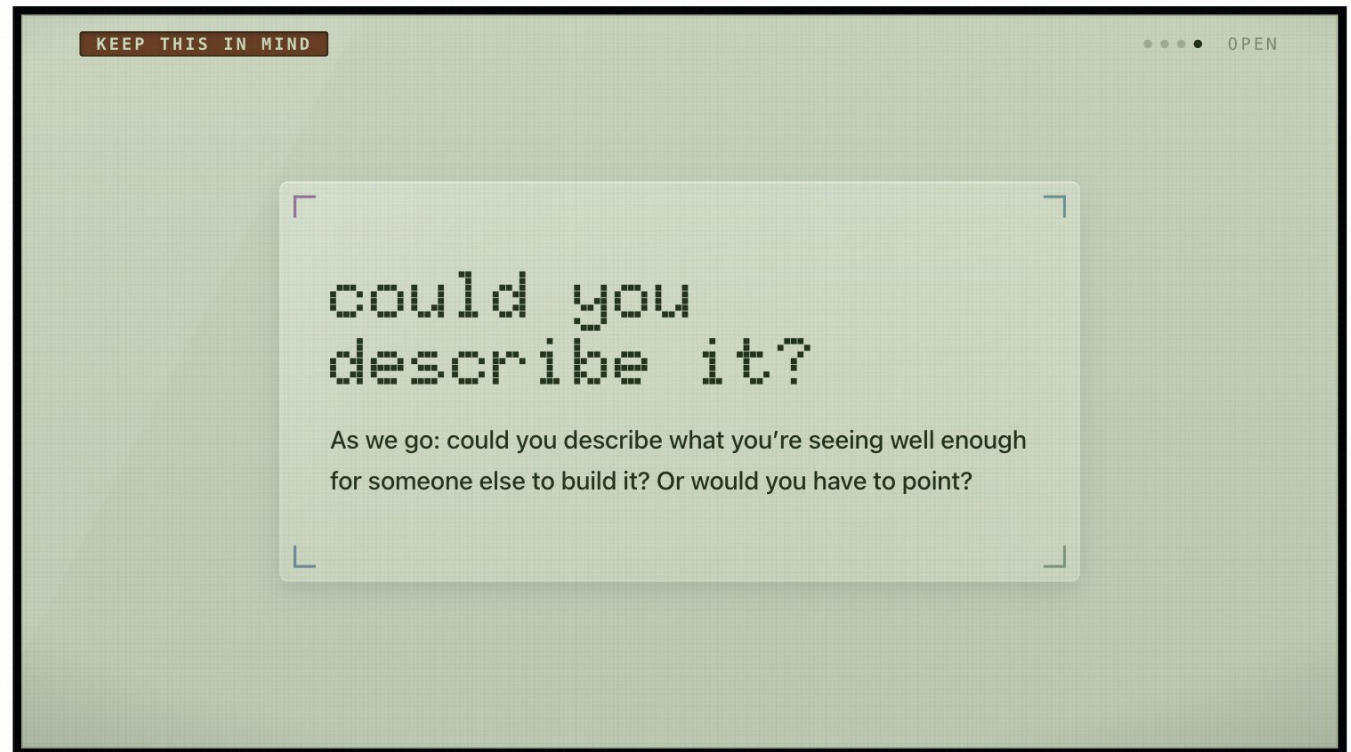
If people stop visiting websites for answers, what's left to visit for is the experience. That's a reading, not a prophecy.

- **Why is an engineer this invested in animation?**
 - *It's not visual interest, it's systems work that happens to be visible*
- **Decoration**
 - A one-off added on top
 - Owned by nobody, inherits nothing, ages out
 - Approved once, as a cost
- **Infrastructure**
 - Something the rest of the work depends on and builds from
 - Keeps paying
- **A well-built website is already a design system**
 - Motion **outside** it is an orphan: no inheritance, no central updates, every sweeping change a hand-edit
 - Motion **inside** it behaves like any token: change once, propagates everywhere
- **Everything later hangs off this**
 - Cost per change, portability, CMS-driven motion, mix and match



A good site is built from a shared kit of parts. Motion built into that kit can be changed once, everywhere. Motion bolted on has to be redone by hand, every time.

- Could I describe this well enough for someone else to build it?
- Would I need to point, or sketch?
- Carry it through all six stops
 - Comes back at the end



As we go: could you describe what you're seeing well enough for someone else to build it? Or would you have to point?

- **You already know this one**

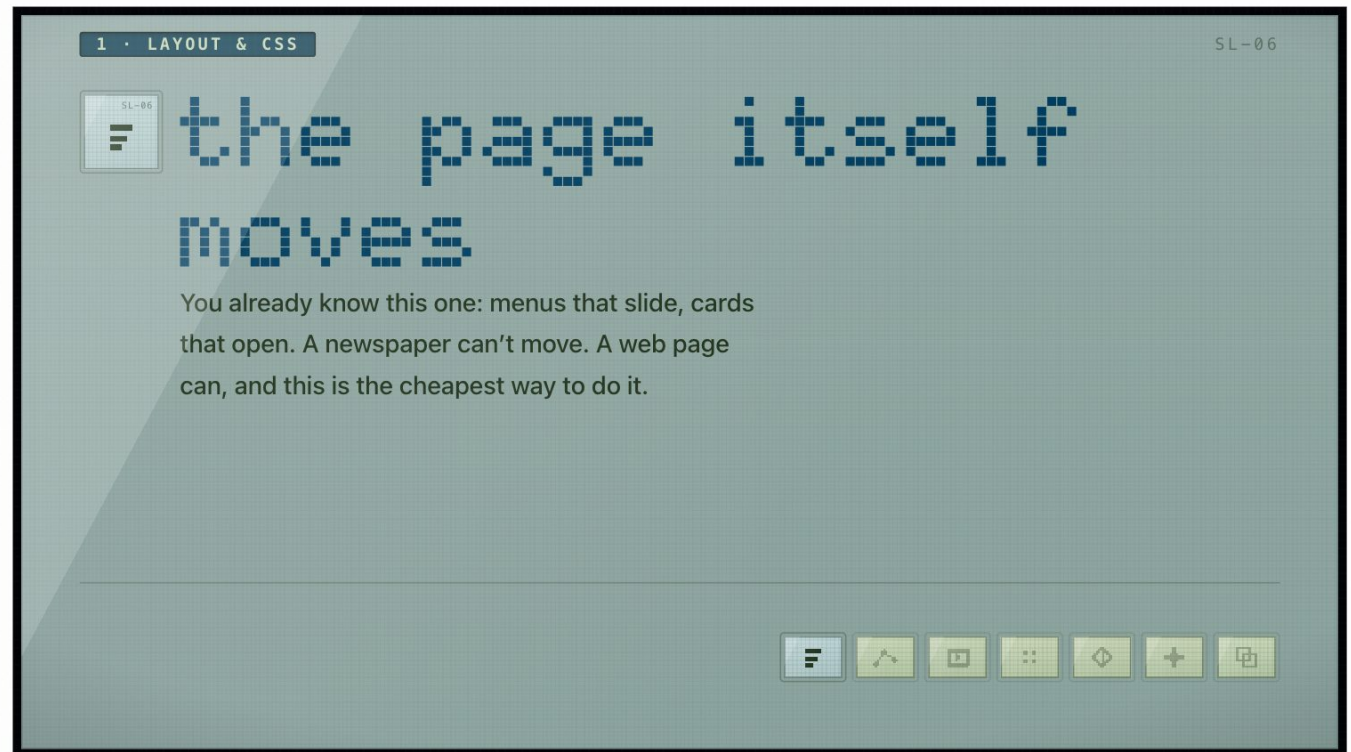
- The most accessible way to add motion to a site
- We already do it, or inherit it through frameworks and kits

- **The print contrast**

- A newspaper is fixed by its medium; a web page isn't
- CSS is the least expensive way to use that difference
- Static content that reads like print → a page that moves

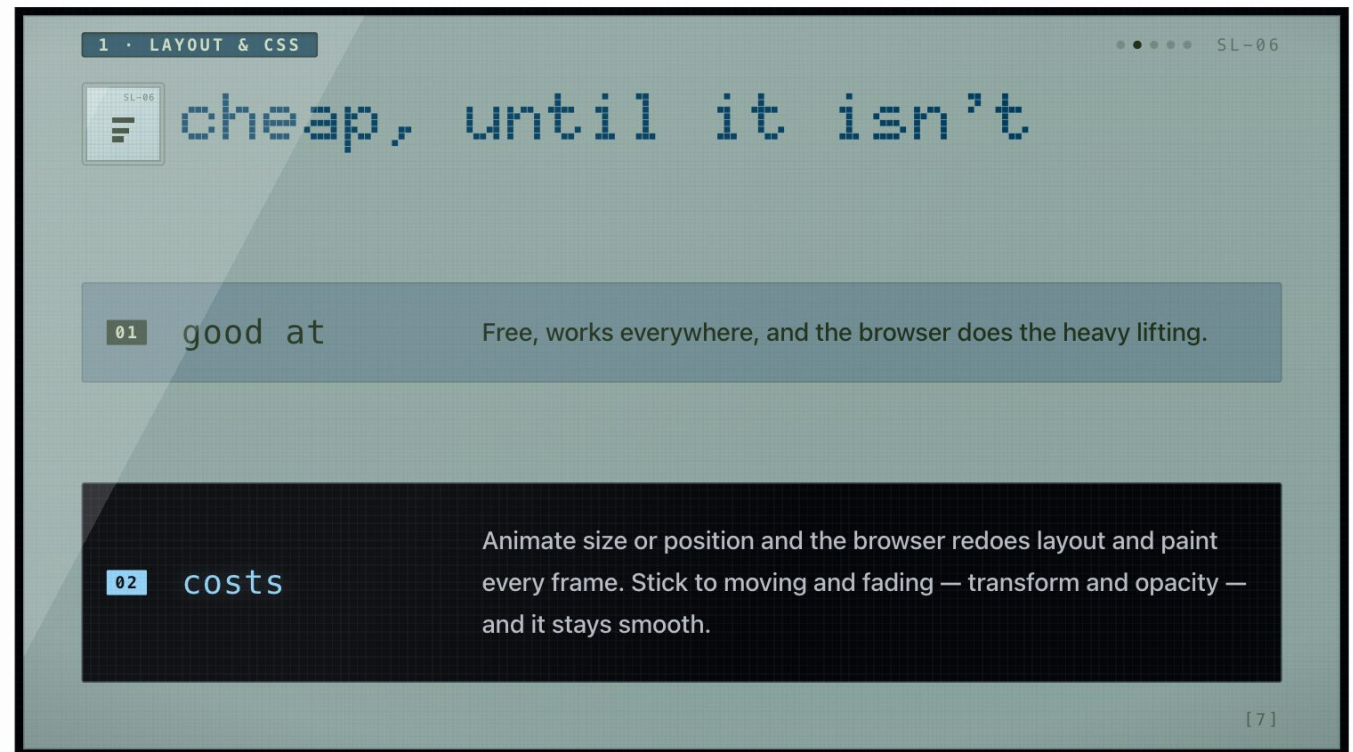
- **Two jobs, one engine**

- **Structure:** *Timeless*, a fictional newsweekly, layout moving with no JavaScript
- **Drawing:** liquid blobs, moiré, holographic type — the part people haven't seen

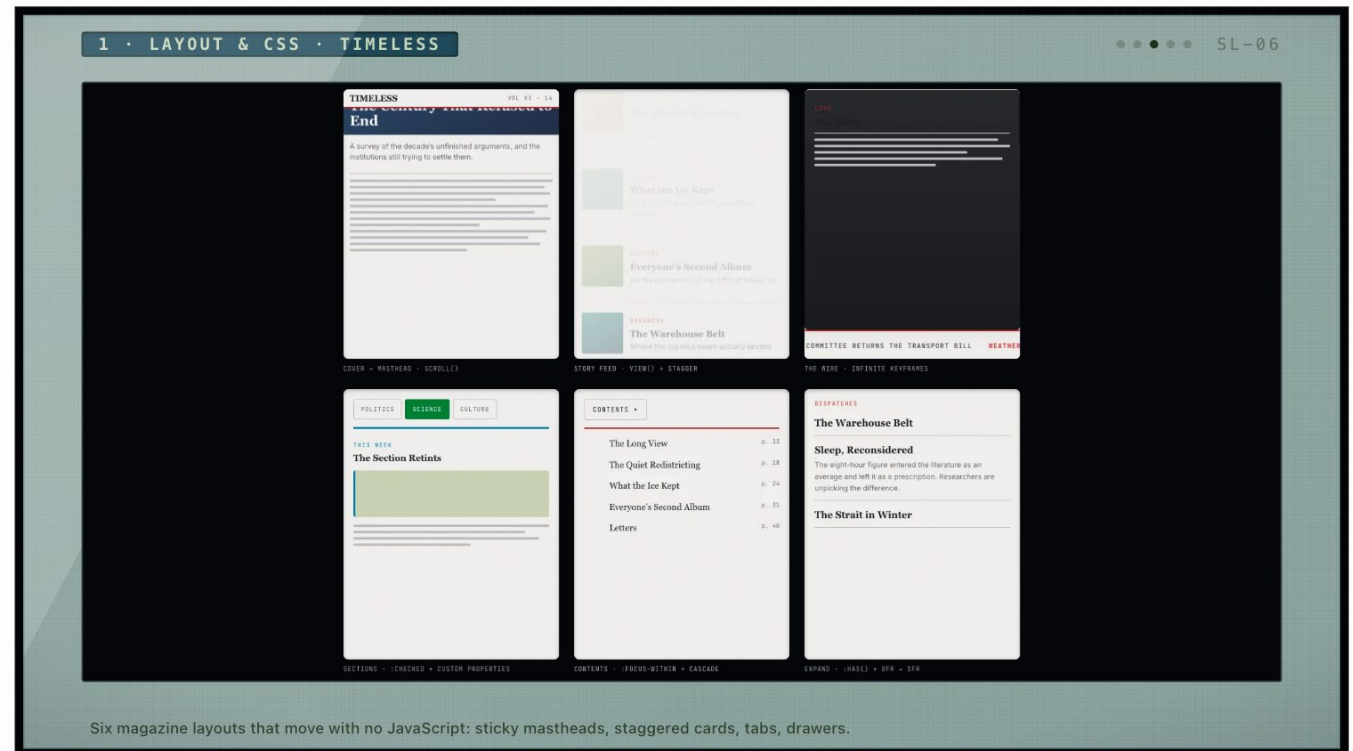


You already know this one: menus that slide, cards that open. A newspaper can't move. A web page can, and this is the cheapest way to do it.

- **Good at**
 - Free, everywhere
 - The only rung where the platform still does the layout work for you
- **Costs: the rendering pipeline**
 - style → layout → paint → composite
 - Animating size or position redoes layout **and** paint every frame
- **transform and opacity**
 - The two properties browsers reliably hand to the compositor [7]
 - Compositor = final assembly: stacks already-painted layers, moves and fades them
 - Each layer is a photo already taken: sliding it is free; resizing means a new photo, and neighbours shuffle
- ***The expensive thing is interrupting the machine's flow***
 - SVG: quantity · Canvas: state switching and overdraw · GPU: asking it a question (readback)
- **The menu everyone has felt**
 - Closed mid-opening, it snaps open, then slides away
 - Reads as **broken, not fast** — "pop-in"
 - Happens with keyframe-built menus; simple transitions already reverse
 - Motion can be built to know where it is, so it just turns around



- **Timeless: a fictional newsweekly, animated with CSS alone**
 - No JavaScript anywhere on these six cards
 - The layout itself moves: this is the "page that moves" from the opening
- **The six cards**
 - **Cover:** the masthead packs down as the page scrolls past
 - **Feed:** story cards fade in on staggered delays
 - **Ticker:** a news crawl moved by transform only
 - **Sections:** tabs recolour the page through a registered custom property
 - **Contents:** the list cascades open, and holds open when focused
 - **Drawers:** accordion rows animate open with grid rows
- **Why it's cheap**
 - The browser does the layout work
 - Motion on **transform and opacity** skips layout and paint [7]



Live demo · [css/timeless](#) — Six magazine layouts that move with no JavaScript: sticky mastheads, staggered cards, tabs, drawers.

- **CSS as a drawing instrument**

- Not layout at all: CSS making images

- **Goo (metaballs)**

- Four circles under `blur(11px)` + `contrast(26)`
- The blur makes them bleed; the contrast snaps the edges back, so they merge like liquid

- **Moiré**

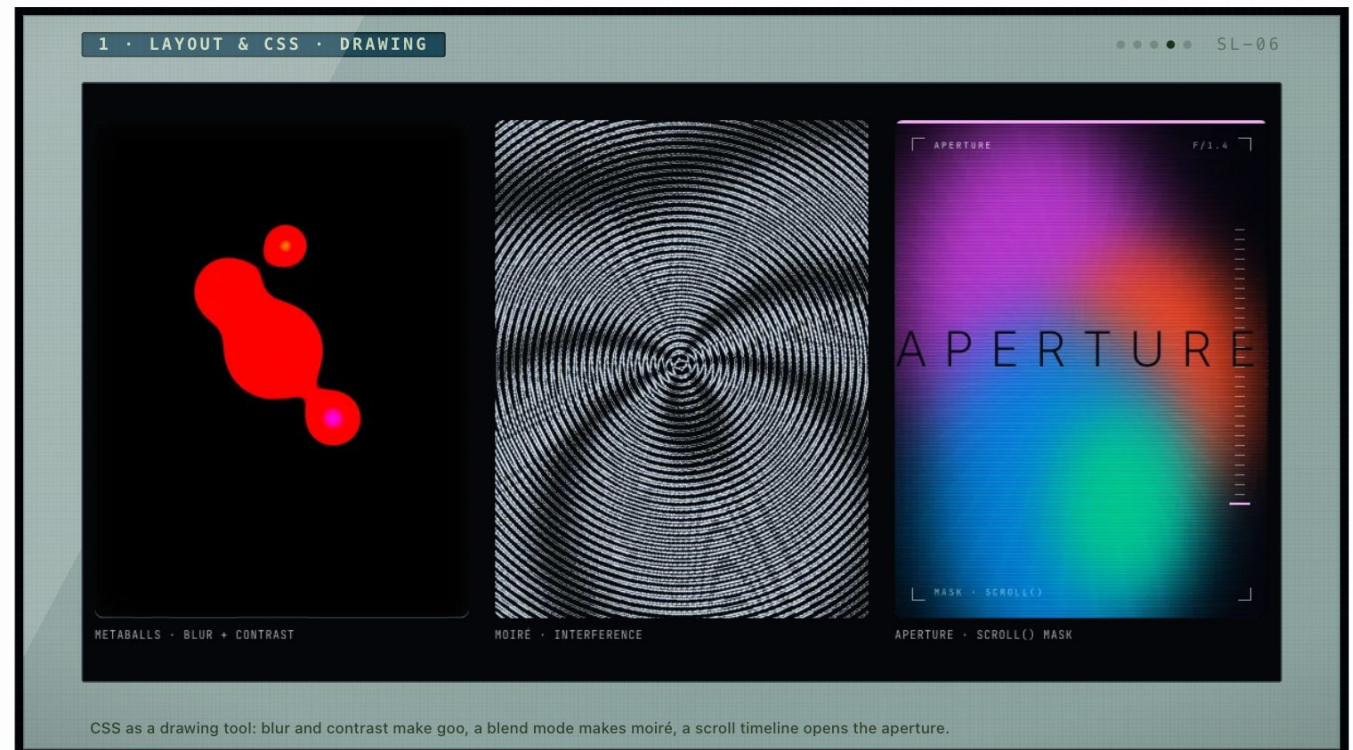
- Two concentric rulings rotating against each other
- `mix-blend-mode: difference` makes the interference pattern

- **Aperture: a scroll-driven mask**

- Scrolling opens a hole in an opaque plate over a drifting colour field
- The field is **revealed**, not moved

- **How**

- The mask radius rides a `scroll()` timeline
- It can animate smoothly only because it's a registered `@property`
- A few lines of JavaScript ease the scrollbar, standing in for a finger; the animation itself is all CSS



Live demo · css/drawing — CSS as a drawing tool: blur and contrast make goo, a blend mode makes moiré, a scroll timeline opens the aperture.

- **Holographic type**

- The word TIMELESS filled with a turning conic gradient
- `background-clip: text` clips the gradient to the letters

- **Layers**

- A ghost layer, a sheen, scanlines, a hollow numeral

- **Still just CSS**

- The type stays real text: selectable, searchable, readable

- **Variable fonts: the one effect that lives inside the font file**

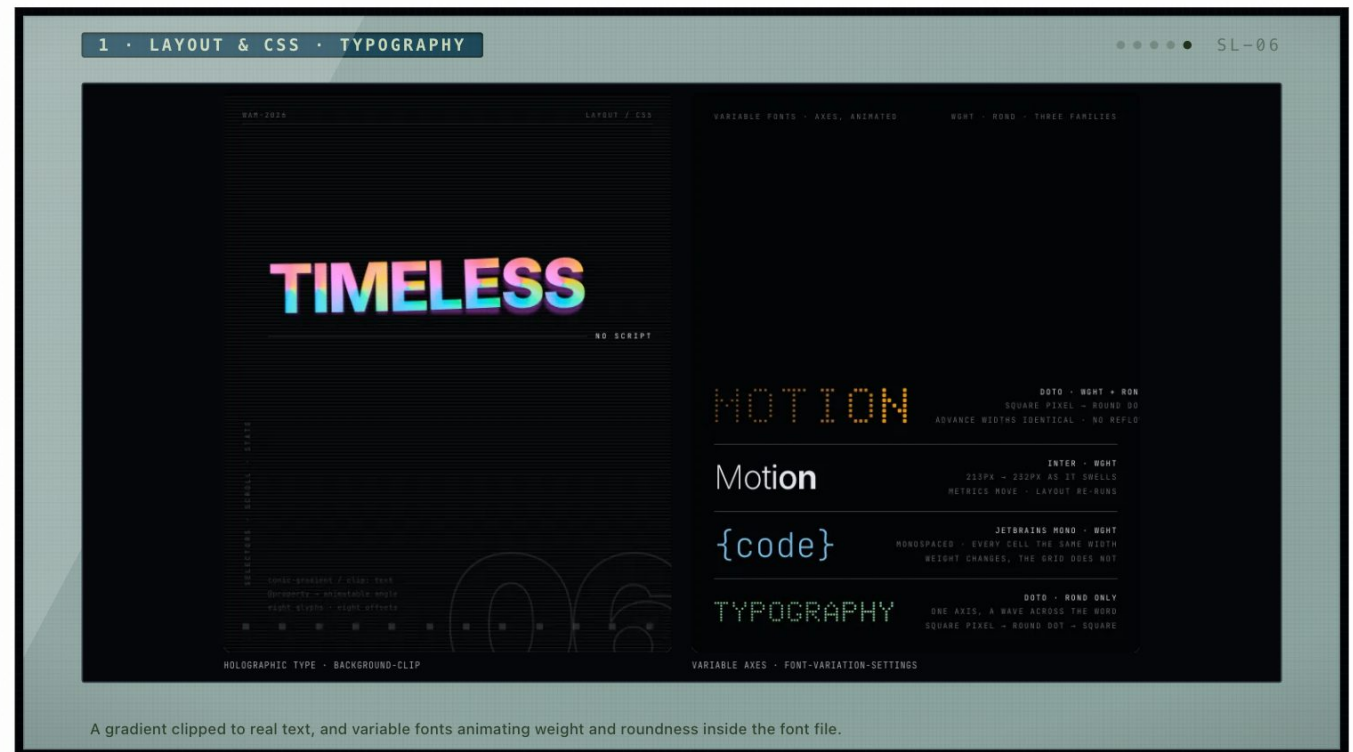
- `font-variation-settings` interpolates the letter **outlines** between axis values

- **Four rows**

- **Doto · wght + ROND:** square pixel to round dot; advance widths never change, so no reflow
- **Inter · wght:** widths grow as it swells, so neighbours shove and layout re-runs
- **JetBrains Mono · wght:** a monospace keeps every cell the same width
- **Doto · ROND only:** one axis, a wave across the word

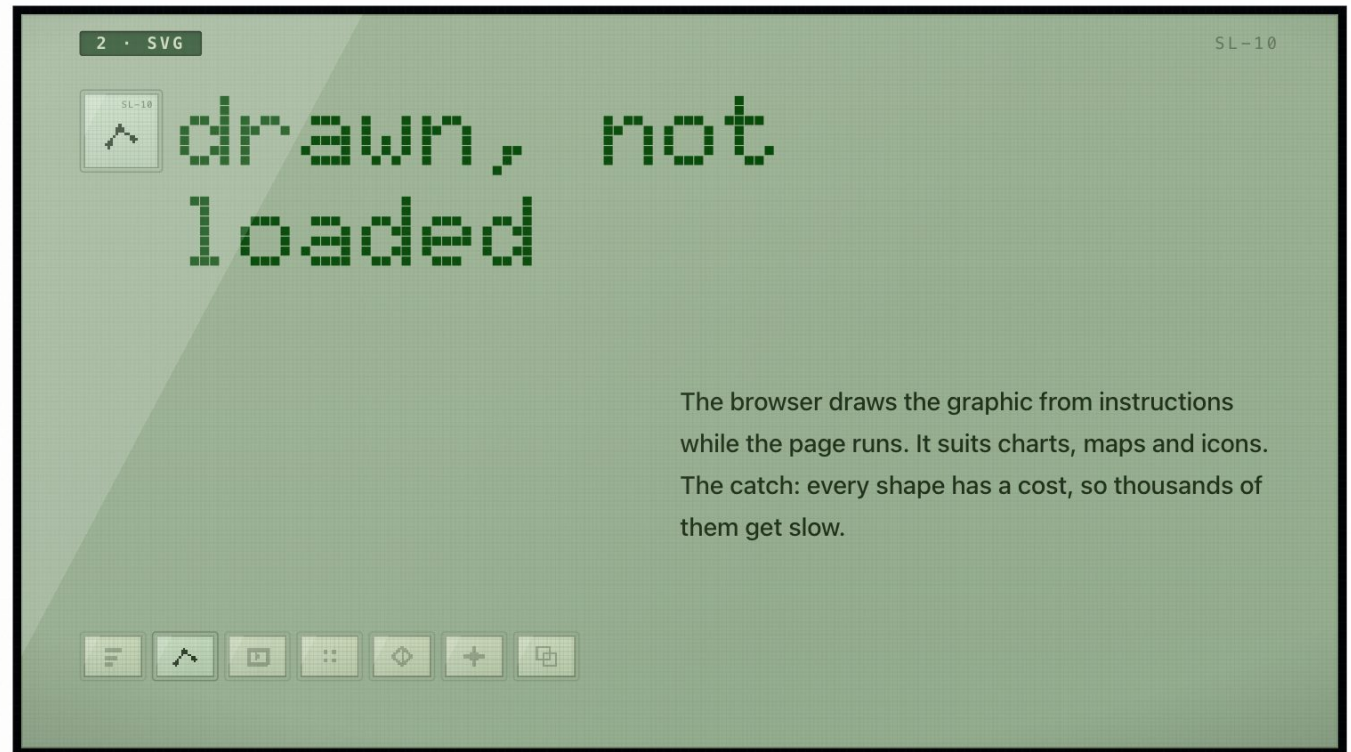
- **Same declaration, different cost**

- Fixed metrics are cheap; changing metrics re-run layout every frame



Live demo · [css/typography](#) — A gradient clipped to real text, and variable fonts animating weight and roundness inside the font file.

- **Graphics as instructions**
 - Not a picture you load; a drawing the browser performs at runtime
- **Where it suits the work**
 - Shapes that come from information, not an artist
 - Data visualisation, maps, small icons
- **Every shape is a real element**
 - Styled, animated and inspected like anything else on the page
- **Costs: quantity**
 - Every node tracked individually → a real ceiling on complexity
 - The graphics card is the way past it (stop 6)



The browser draws the graphic from instructions while the page runs. It suits charts, maps and icons. The catch: every shape has a cost, so thousands of them get slow.

- *CSS animates what's there. SVG lets you define what's there.*
- **CSS**
 - Moves elements that already exist: boxes, text, layout
- **SVG**
 - The shape itself is what you describe
 - Arbitrary forms, changed point by point
- **Why people conflate them**
 - A simple shape looks identical either way
 - The difference shows when the shape comes **from data** or **morphs** into something else

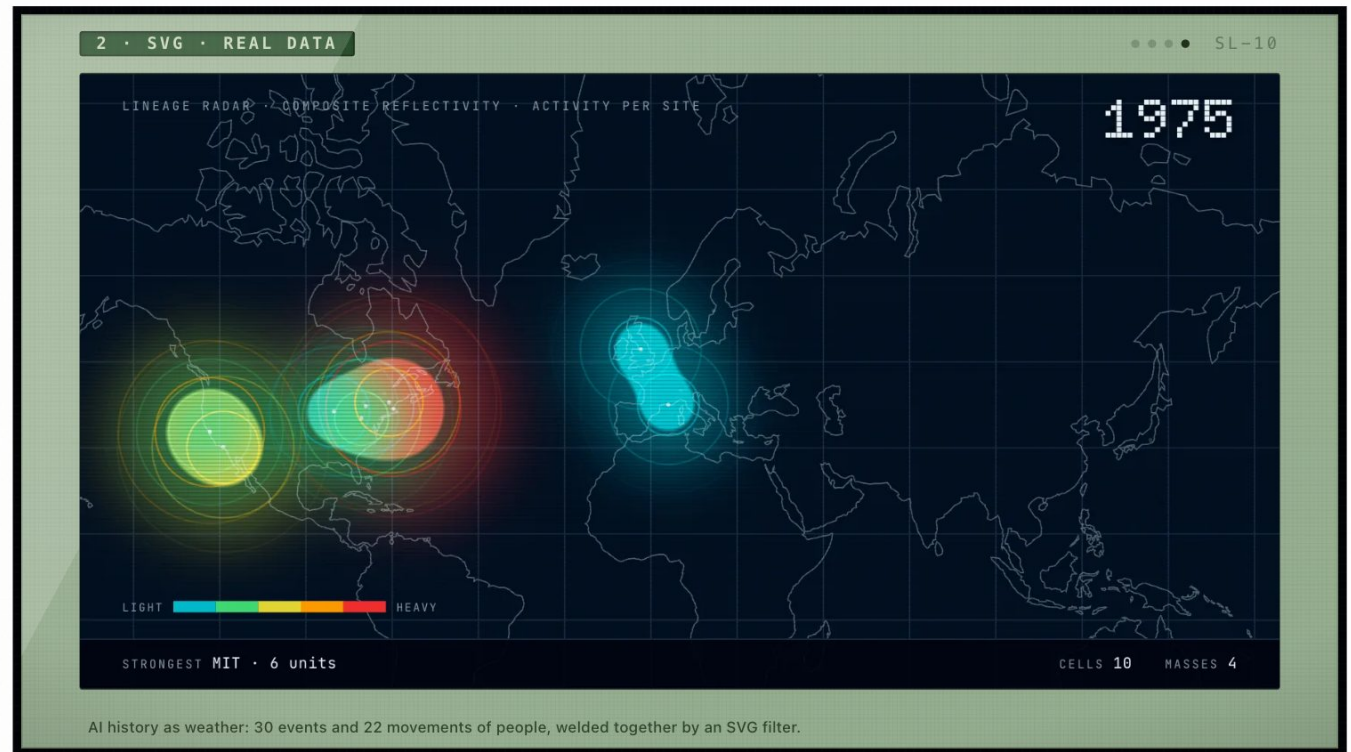


- **A survey sheet, zooming from the world to Dartmouth College, 1956**
 - Where the Dartmouth workshop named "artificial intelligence"
- **The camera is SVG**
 - One nested group transform moves the whole map; the browser redraws the vector at every scale
 - Coastlines swap to finer data as you zoom; borders and labels crossfade in
- **Why SVG suits it**
 - Shapes from data, not pixels: sharp at any zoom
 - Labels are real text, placed and culled so they never collide
- **The cost**
 - Every path and label is a tracked element; that's the ceiling



Live demo · [svg/zoom-map](#) — One SVG camera flies from the world to Dartmouth College, 1956: sharp at every zoom, labels as real text.

- **Lineage radar: AI history as weather**
 - 30 events and 22 sourced movements of people between institutions, 1943–2024
 - Each site's mass grows with the events and movements on record there by that year
- **Melding and radiating**
 - An SVG filter (blur, then a hard threshold) welds nearby masses into one shape
 - Rings pulse out from every active site
- **Honest data**
 - Every size is counted from the dataset, not tuned for the picture
- **Still SVG**
 - A filter chain the browser runs, no shader code



Live demo · [svg/lineage-radar](#) — AI history as weather: 30 events and 22 movements of people, welded together by an SVG filter.

- **The format**

- Just works, everywhere, no code — drop in an `<img>`
- Pre-baked pixels, **no controls**: can't pause
- Can't honour reduced motion without swapping the file
- 256-colour palette → gradients band

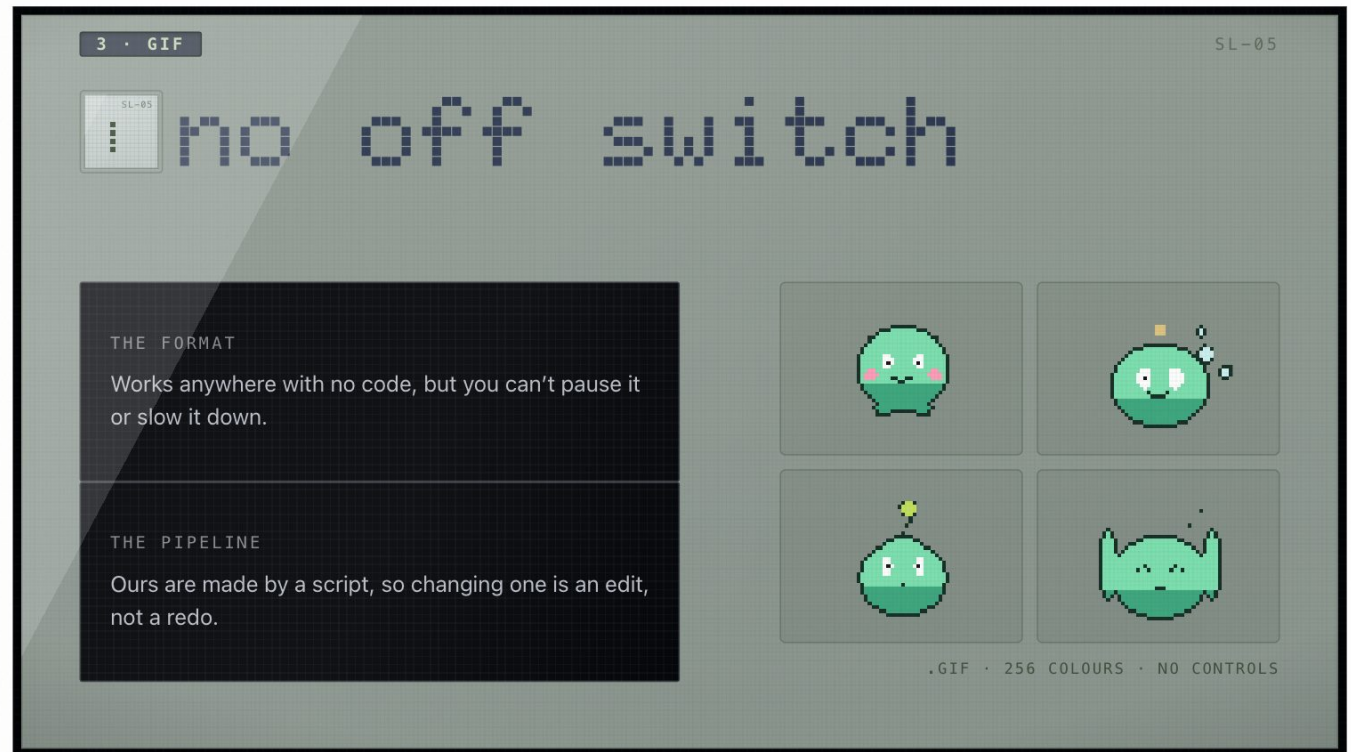
- ***The one format that takes the decision away from you***

- **The pipeline — the part people miss**

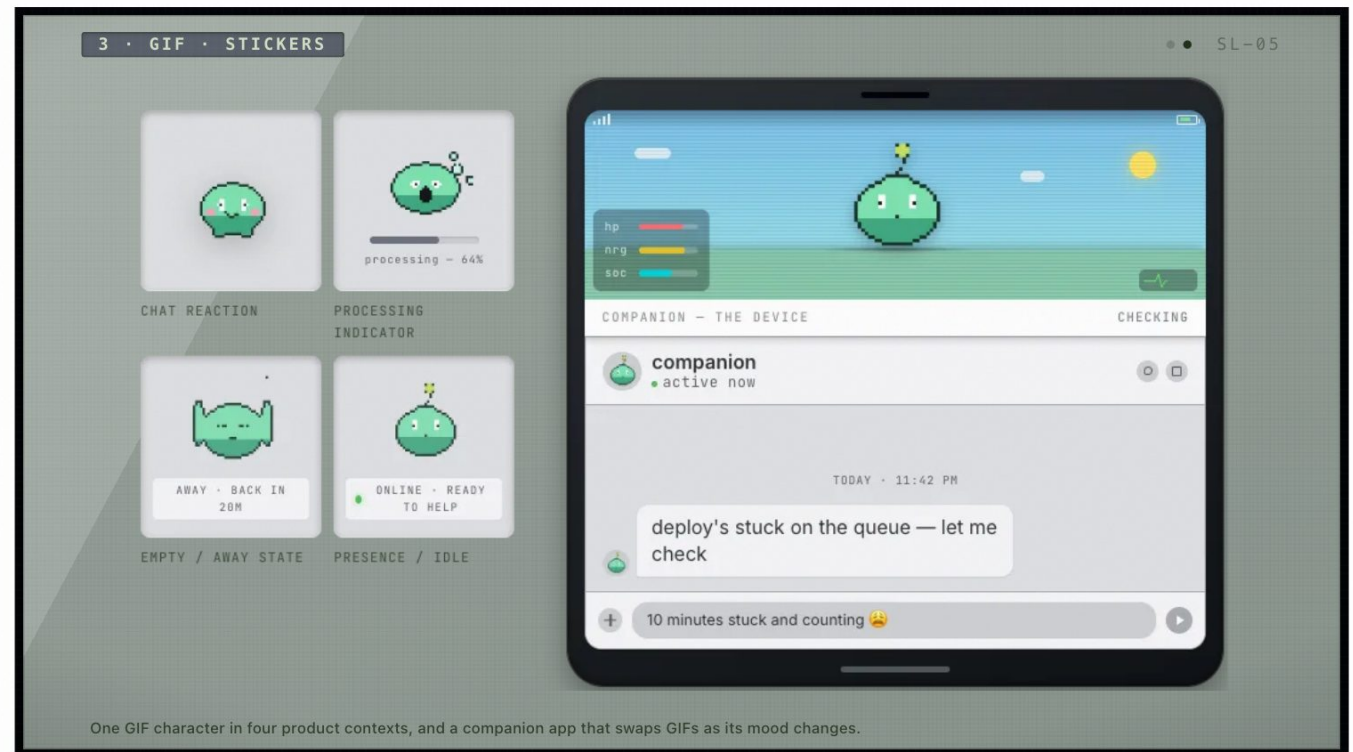
- Ours come out of a rendering pipeline: change a value, re-bake
- Not a file someone made once
- Even a GIF can be an **output**, not an artefact

- **Client angle**

- Existing video, 3D or motion can become GIFs used as UI, not memes



- **GIFs as UI, not memes**
 - One slime family, staged in real product contexts
- **Four contexts**
 - **Chat reaction:** an oversized sticker in a conversation
 - **Processing:** a GIF above a progress bar
 - **Away state:** a sleeping GIF with an "away" tag
 - **Presence:** the idle loop as an "online" mark
- **The companion device**
 - A scripted chat swaps the character's mood as the conversation plays: idle, eat, happy, sleep
 - Each mood is just a different GIF file
- **The format**
 - Works everywhere, no code; can't be paused or honour reduced motion without swapping the file
 - 256 colours, so gradients band
- **The pipeline**
 - Generated by a script: change a value, re-bake



Live demo · gif/stickers — One GIF character in four product contexts, and a companion app that swaps GIFs as its mood changes.



- **Good at**
 - Cheapest way to put motion on a screen
 - The only pre-baked format you can tell to stop
- **Costs: it's fixed**
 - **Video can't respond** — not to a cursor, scroll or click
 - Once motion has to respond, video isn't cheaper; it isn't an option
- **Three uses we haven't tried, and could**
 1. **Filters on captured video** — effects on a live camera feed, Instagram-style
 2. **Computer vision** on video or live streams — adjacent, not shown
 3. **Video as a component** — keyed footage layered so it reads as part of the page
- **The video is not the content, it's a material**
- **Cost per change**
 - Static asset: every revision = open source, re-render, re-export, re-upload
 - Each round costs roughly what the first did
 - Living system: the same request is a value edit and a reload
- **The cost is payroll, not invoice**
 - Designer hours + coordination × rounds, hidden in salaried time
 - Scotty Cameron's 3D club animation: every revision round was a redo
 - One-off asset, or a system that generates assets?
- **Portability**
- The render target is a detail, not the project
- Live page · HD video · YouTube cut · vertical social spot · broadcast overlay **[14]** · native
- **The After Effects contrast**
 - A rendered file is inert; the intelligence is gone at export
 - Shader and canvas work stays live
 - **You're not producing videos, you're producing something that emits them**
- **CMS-driven motion**
 - Code motion is values → values can be CMS fields
 - Client changes a colour, headline or speed → animation updates; no re-render, no ticket
 - A video in a CMS is just a file

- **NOVA-7: a fictional handheld**
 - Baked by a custom software renderer: depth buffer, per-pixel Fresnel, 3x supersampling, a loop with no seam
- **The decoded frame is a buffer**
 - Once video is decoded, every frame is pixels you can read and rewrite
 - The rack runs one effect at a time over the playing video
- **The rack**
 - **Buffer:** slit-scan, ripple, datamosh, mosaic drift, feedback, kaleidoscope, drawn from a frame history
 - **Filters:** edge detect, chroma split, bloom
 - **Composite:** mirror, echo, difference, using a second decoder offset in time
 - **Time:** drive playback from an easing curve
 - **Camera:** the same rack on a live camera feed
- **Why it matters**
 - Filters on captured video: a different product from playback, same technique family
 - *The video is not the content, it's a material*



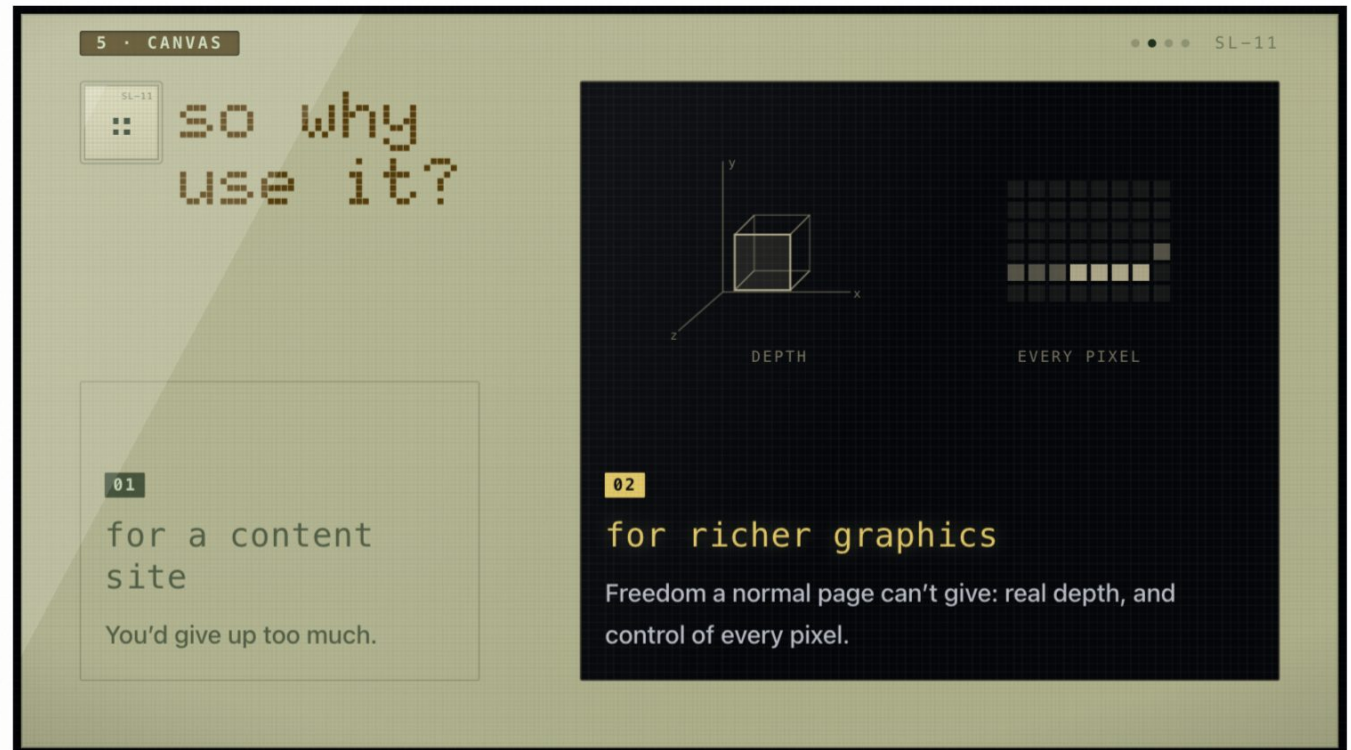
Live demo · video/player-rack — The decoded frame is a buffer: effects rewrite a playing video, or a live camera, frame by frame.

- **Try to select the text**
 - A test anyone can run on any site
 - Normal page: highlight text, drag an image, tab to a link
 - Canvas: none of it works — looks like content, nothing to grab
- **Why: no scene graph**
 - Nothing drawn is remembered — no element, no node
 - Just pixels you put there and manage yourself
- **Costs: everything the platform did stop**
 - No layout engine, no accessibility tree, no handles for tests
- **Responsiveness without layout**
 - Canvas resizes; content doesn't reflow
 - Think **film framing, not web layout**: compose for the extremes, keep what matters in a safe area
- **As you climb the ladder, the platform stops doing things for you**

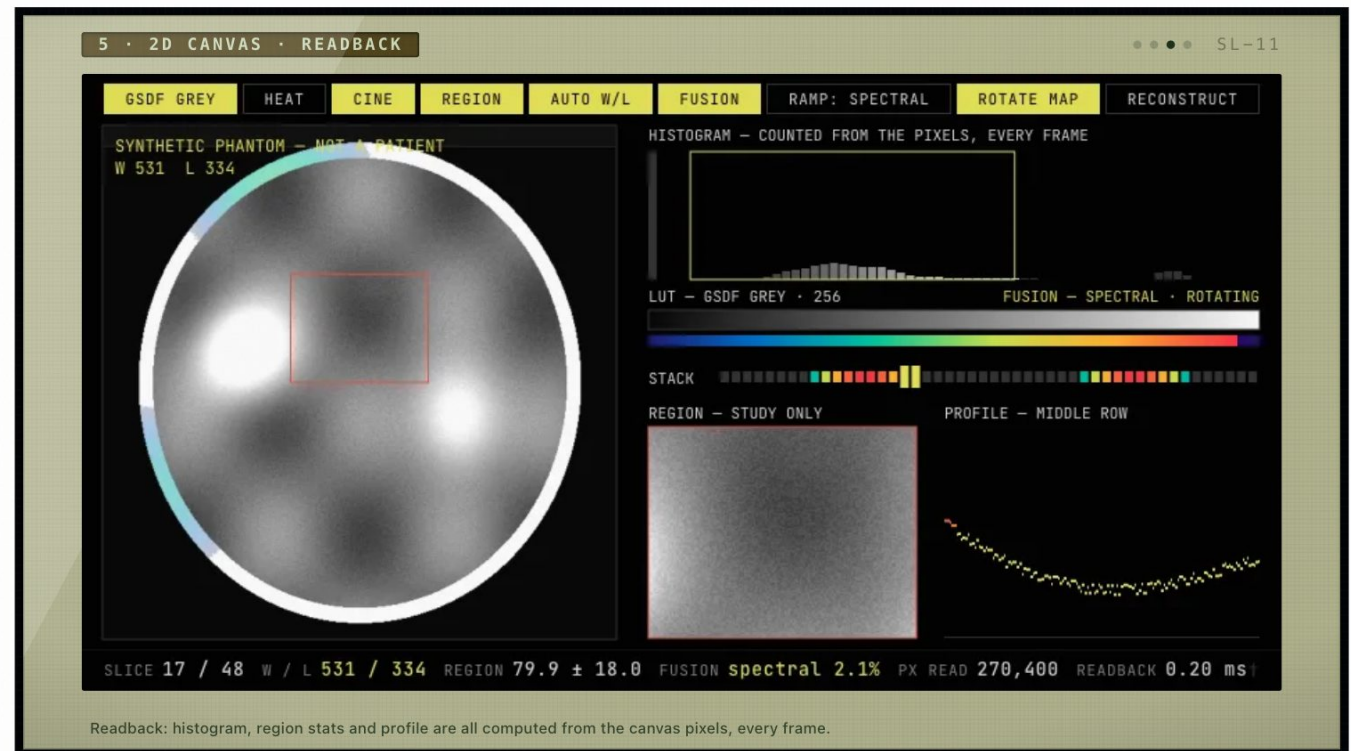


On most sites you can highlight text, drag an image, tab to a link. On a canvas you can't. It's just pixels the page painted, and it remembers nothing.

- **So why choose it?**
 - Freedom the layout system can't give
- **What you get**
 - Work in the **Z axis** — real depth
 - Direct access to the **pixels** — what filters and advanced effects need
- **The trade**
 - Typical content site: a **huge loss**
 - Extending graphics capability: the **better option**
- **Hand-off**
 - WebGL and WebGPU are ways of drawing onto this same surface



- **Readback: the answer is in the pixels**
 - A synthetic 48-slice scan, not a patient
- **Everything on the right is computed from the image, every frame**
 - Histogram, region stats, the magnified study, the profile line
 - Change the slice, window, level or colour map and all of them follow
- **Why canvas**
 - A layout engine can tell you where its objects are
 - It can't answer these, because the answer isn't in the objects, it's in the pixels



Live demo · canvas/readback — Readback: histogram, region stats and profile are all computed from the canvas pixels, every frame.

- **Off-thread: two identical players**
 - Unit 1 is drawn on the page's main thread
 - Unit 2 is drawn by a worker through `OffscreenCanvas`
- **Block the main thread for 800 ms**
 - Unit 1 freezes; its scrolling title stops
 - Unit 2 keeps running
- **Same pixels**
 - A parity check proves both draw byte-identical frames
- **The point**
 - Canvas drawing can leave the main thread; the page stays responsive



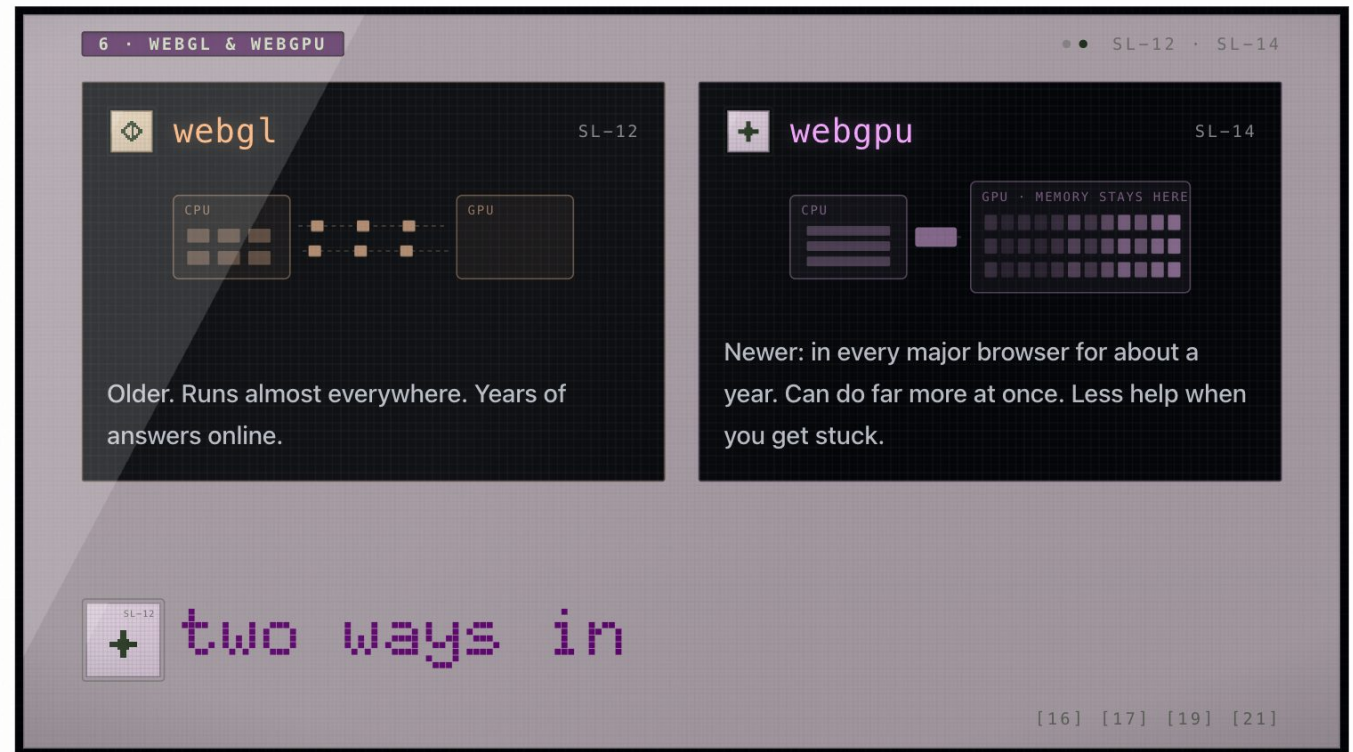
Live demo · canvas/off-thread — A worker draws the player and visualizer through OffscreenCanvas; block the page and they keep running.

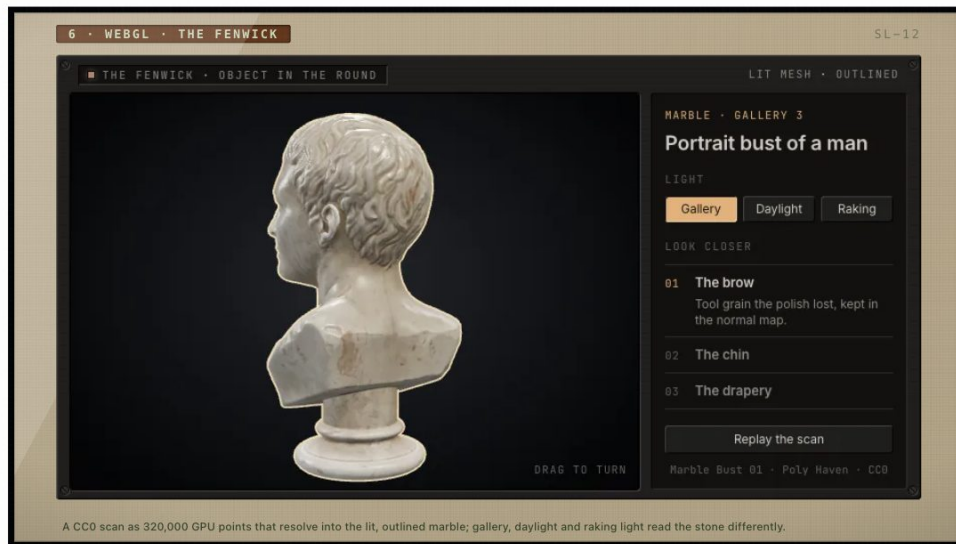
- **Think of a video game you've played**
 - Or an interactive experience outside a web page
 - Much of the difference from a website is the graphics
- ***That gap is the graphics card***
 - WebGL and WebGPU let a web page use the same chip that runs games
- **An enormous amount of small work, all at once**
 - A flock of **131,072 birds** instead of a dozen
- **Two questions decide which**
 - How much simulation or calculation do you need?
 - How much support and community do you want to lean on?



Think of a video game, then think of a website. Much of that difference is the graphics card, and these let a web page use it.

- **WebGL**
 - Older; OpenGL ES in a web coat [15]
 - Runs almost everywhere
 - Fifteen years of answers — final 1.0 spec in 2011 [16]
 - Not limited to static: models, lighting, effects are comfortable
- **WebGPU**
 - Newer; can do more at once — compute on the GPU
 - Chrome 113, 2023 [17] · Safari 26, Sept 2025 [19] · Firefox 141 Windows, 147 Apple Silicon Macs, Jan 2026 [20] [21]
 - Thinner ecosystem; less help when you get stuck
- **Use WebGPU where support allows, WebGL as fallback**
- **2D vs 3D**
 - Same machinery either way
 - 2D on the GPU is the escape from **SVG's quantity ceiling**
- **Cost**
 - *Unpredictable in advance, measurable once built*
 - **Name the target device first**
- **Tooling**
 - Real progress on enablers — e.g. Vercel's `vgpu`, public 27 Aug 2026 [34]





Live demo · [webgl/fenwick](#) — A CC0 scan as 320,000 GPU points that resolve into the lit, outlined marble; gallery, daylight and raking light read the stone differently.

- **The Fenwick: a museum's object page (invented)**

- A real photogrammetry scan: Marble Bust 01, Poly Haven, CC0

- **320,000 points of light**

- Sampled from the scan's surface; they gather into the bust in about two seconds
- The points live on the GPU; nothing is uploaded per frame

- **Points → object**

- A scan line rises; below it the points give way to the lit mesh

- Both are drawn from the same mesh in the same space, so they line up exactly

- **Outline**

- A second copy of the mesh, turned inside out and pushed a pixel or two past the edge
- It separates pale stone from a dark ground without any post-processing

- **Light as content**

- Gallery, daylight and raking light: the same stone reads differently under each

- The fine tool grain is in a normal map, not the geometry, so it turns with the light

- **Look closer**

- Each note turns the bust to show the detail

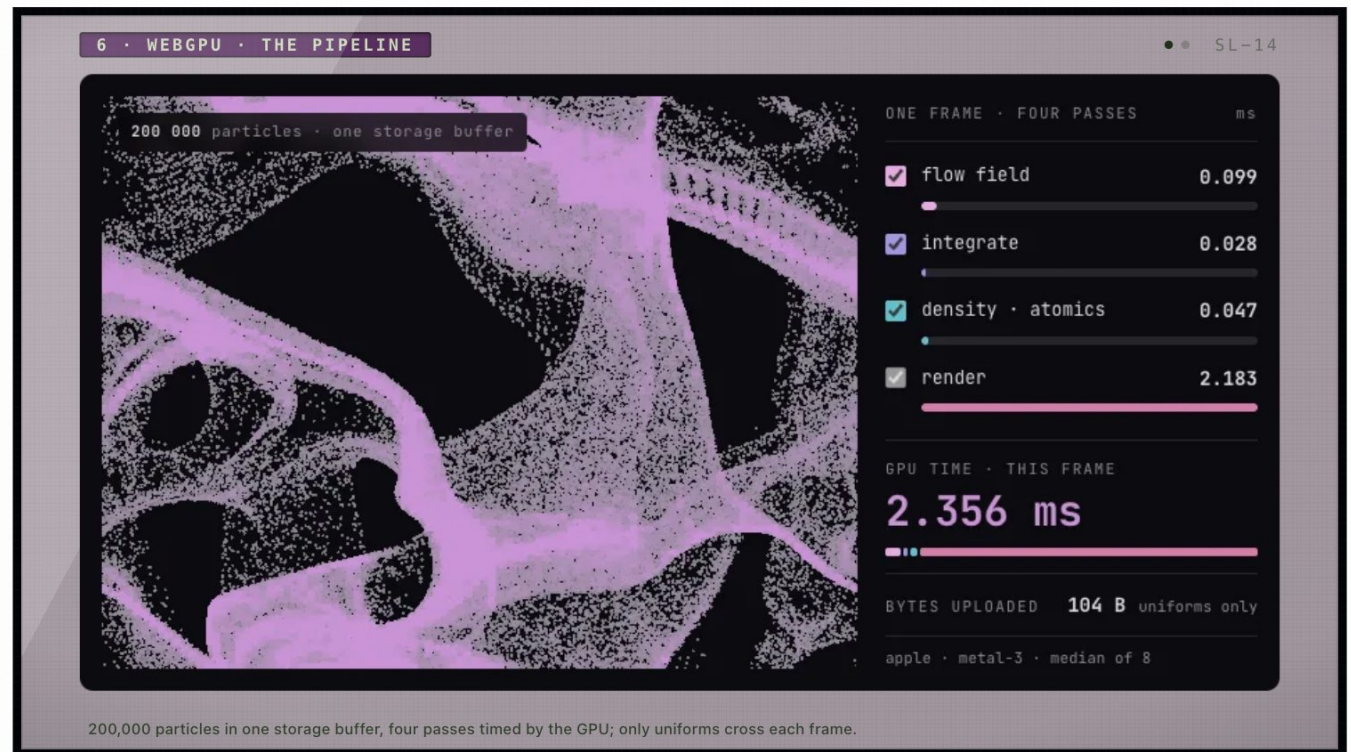
- **A composite**

- Real text and buttons where they need to be read; the graphics card where it needs pixels

- **WebGL is not only for static scenes**

- Models, lighting and effects are comfortable; its limit is lots of interacting state

- **What a frame costs on the GPU**
 - 200,000 particles, one storage buffer
- **Four passes**
 - Flow field, integrate, density, render
 - Each pass is timed by the GPU itself, start and end
 - Switch a pass off and the frame loses exactly its time
- **Bytes uploaded: uniforms only**
 - The particles never come back to the CPU; only a few parameters cross each frame
- **Why WebGPU**
 - Compute shaders: the GPU holds the state and evolves it



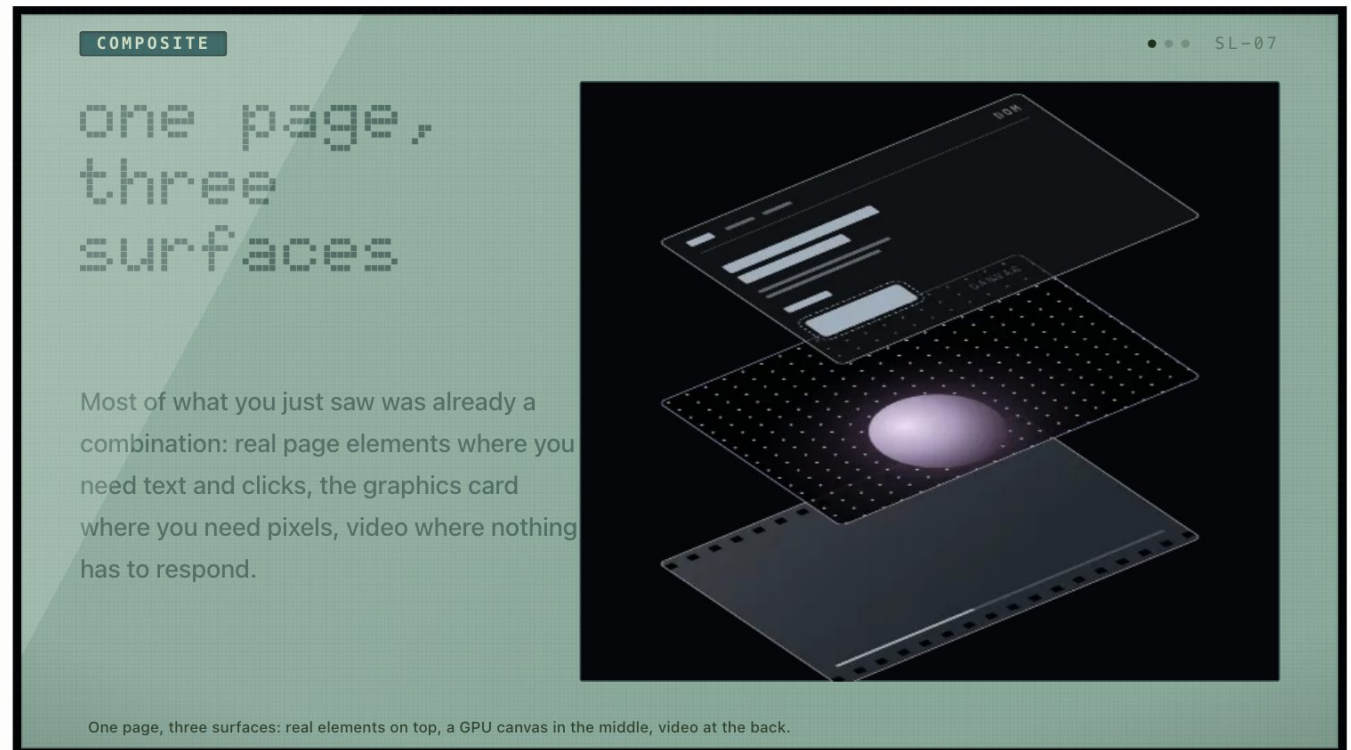
Live demo · [webgpu/pipeline](#) — 200,000 particles in one storage buffer, four passes timed by the GPU; only uniforms cross each frame.

- **One device, one canvas, all WebGPU**
 - The fPhone Duo itself is rendered on the GPU; the home screen, widgets and apps are drawn inside it
- **The apps**
 - **Roost:** 131,072 starlings, each reading its six or seven nearest neighbours [29]
 - **Tessera:** a world grown from noise and erosion, ray-marched with light, shadow, water and fog
 - **Supercell:** a storm that runs itself
- **One app alive at a time**
 - Opening an app tears down the last one; a single set of buffers is ever alive



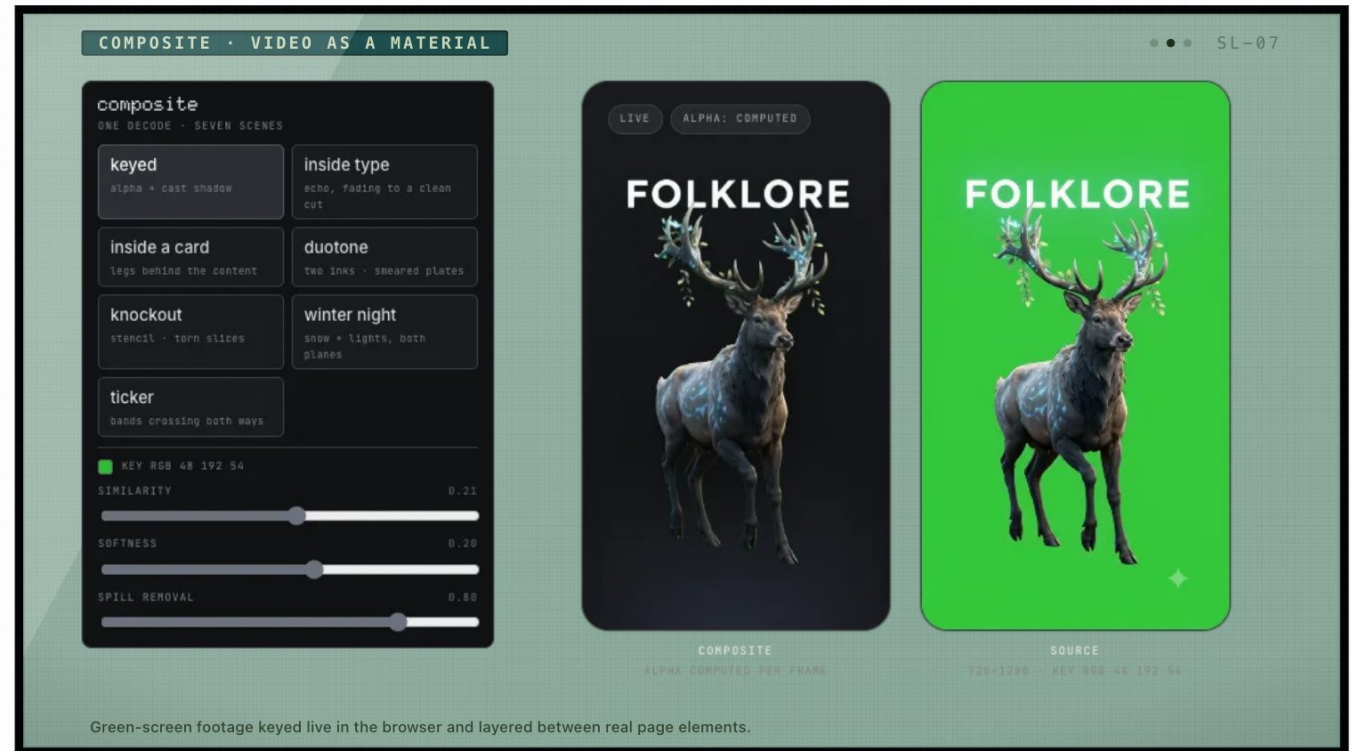
Live demo · [webgpu/device](#) — One canvas, all WebGPU: the fPhone Duo, its home screen and every app drawn by the graphics card.

- **Most of what you just saw was already a combination**
 - Real page elements where you need text, clicks and accessibility
 - The graphics card where you need pixels
 - Video where nothing has to respond
 - An alternative for anyone who can't use the GPU layer
- **One page, three surfaces**
 - The user sees one page; the browser runs three layers
 - A canvas is one opaque rectangle to the page, so nothing that must be read goes in it
- **Composites are a performance strategy, not just an aesthetic one**
 - Each part on the cheapest surface that can carry it
 - Keep the parts that move out of the expensive pipeline stages



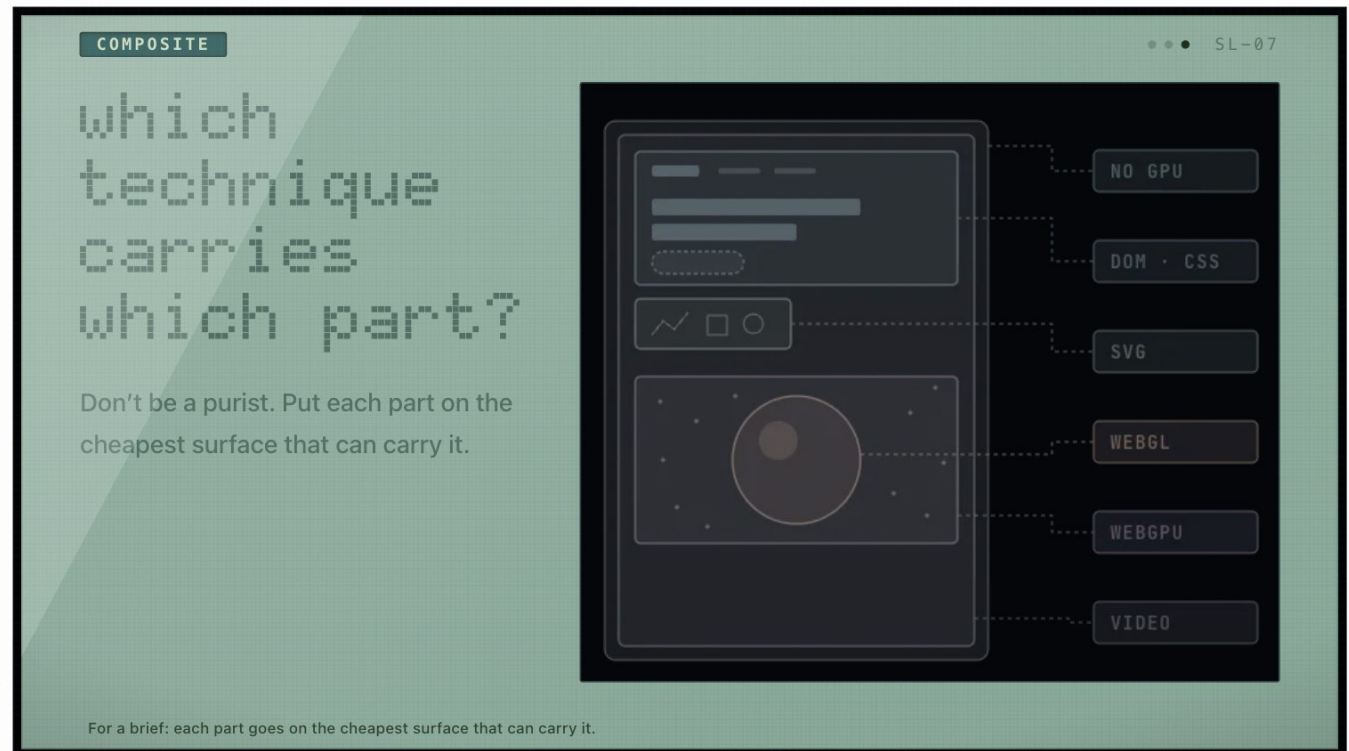
Live demo · composite/layers — One page, three surfaces: real elements on top, a GPU canvas in the middle, video at the back.

- **An alpha channel the file never had**
 - Footage shot against green screen, keyed live in the browser
 - What remains is layered between real page elements, in front and behind
- **Seven scenes, one decode**
 - Float, inside type, inside a card, duotone, knockout, winter night, ticker
- **Video as a component**
 - Not a YouTube embed, not a cinematic piece: a material placed into the page
 - The page elements stay real: selectable, accessible, editable
- **A composite**
 - Video carries the motion, the page carries the text, each on the surface that suits it



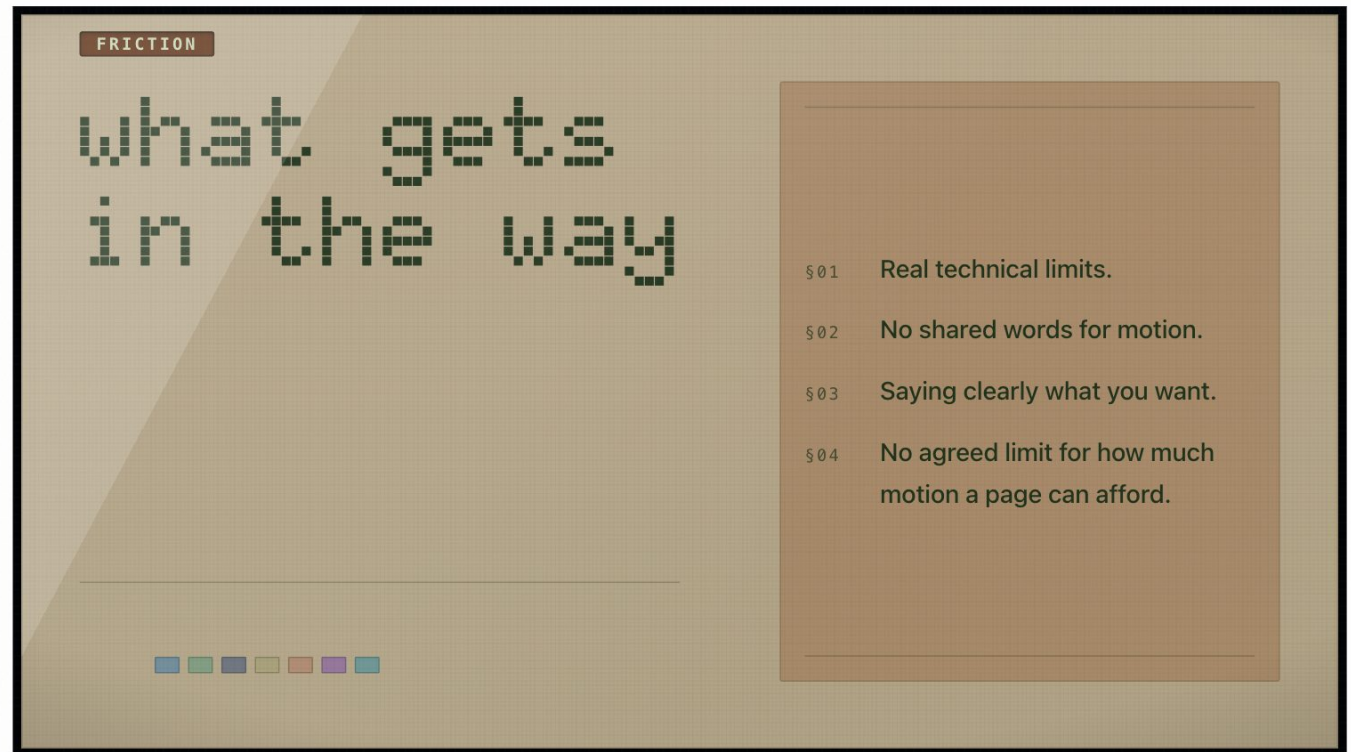
Live demo · [composite/chroma-key](#) — Green-screen footage keyed live in the browser and layered between real page elements.

- **Don't be a purist**
 - Purism forces one surface to do work it's bad at
- **For a brief**
 - Not "which technique is this project"
 - **Which technique carries which part?**
 - The failure mode: a brief that assumes one answer before anyone has looked
- **Reading the picture** (*illustrative launch page; one part lights at a time*)
 - Nav, headline, buy button → page elements (DOM · CSS)
 - Line icons → SVG
 - Hover and state changes → CSS, **transform and opacity** skip layout and paint [7]
 - Ambient loop → video, until it has to respond
 - The product under the cursor → WebGL; a field of interacting particles → WebGPU
 - No GPU → a still, with the same page elements on top



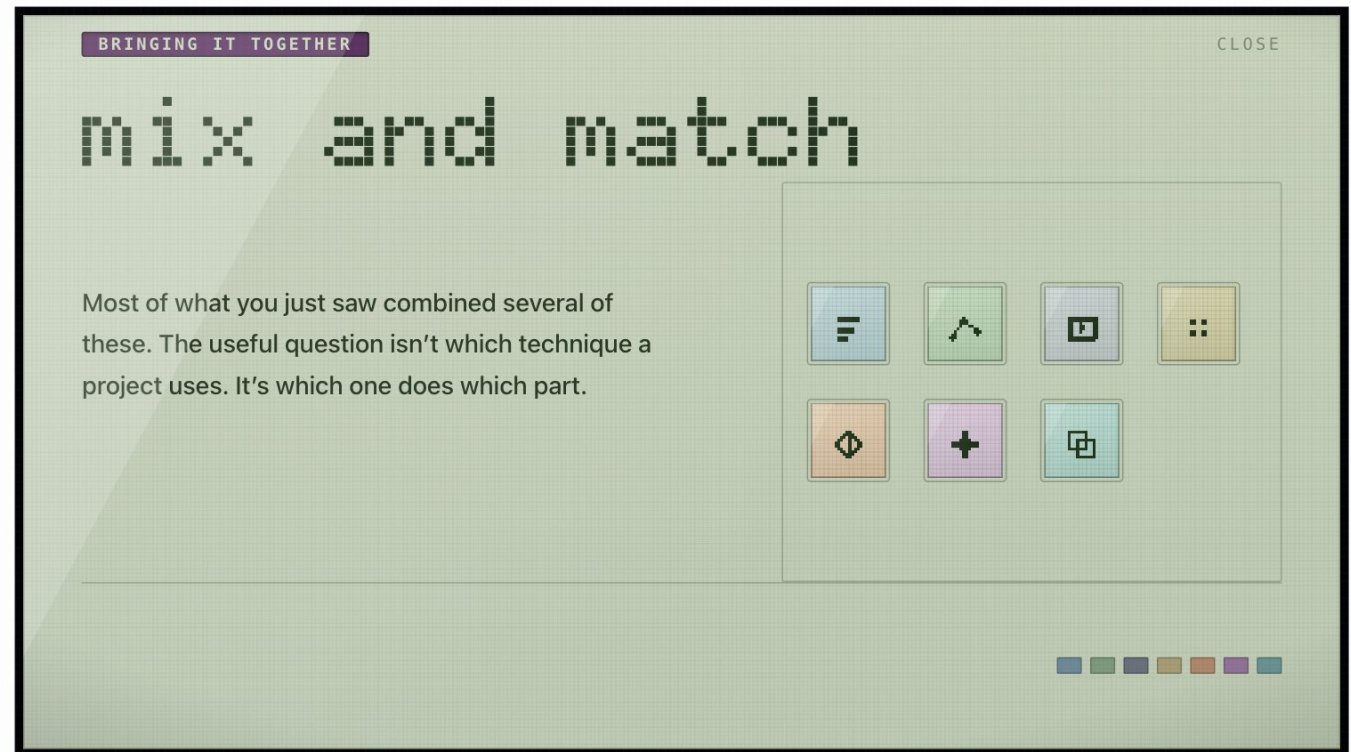
Live demo · composite/carries — For a brief: each part goes on the cheapest surface that can carry it.

- **1 · Technical constraints**
 - Real, and engineering's to own
- **2 · Shared vocabulary**
 - A menu can be shown in two stills; a fluid simulation can't
 - Words fill **exactly the gap the design tool can no longer show**
- **3 · Naming what you want**
 - With AI building, speed tracks how precisely you can name it — not code complexity
- **4 · No performance budget for motion**
 - Web vitals work because someone picked a line: main content in **2.5 s [43]**
 - Arbitrary, but **shared** — so actionable
 - Motion has no equivalent: feel, dropped frames, a hot device
 - Without a number, every request is argued on vibes
 - ***Give engineering a number and refusal stops being the safe default***
 - Naming the target device is a **product** decision
- **The tooling gap**
 - Free tools exist, built for a developer alone, after the fact
 - ***Not access, it's design intent***



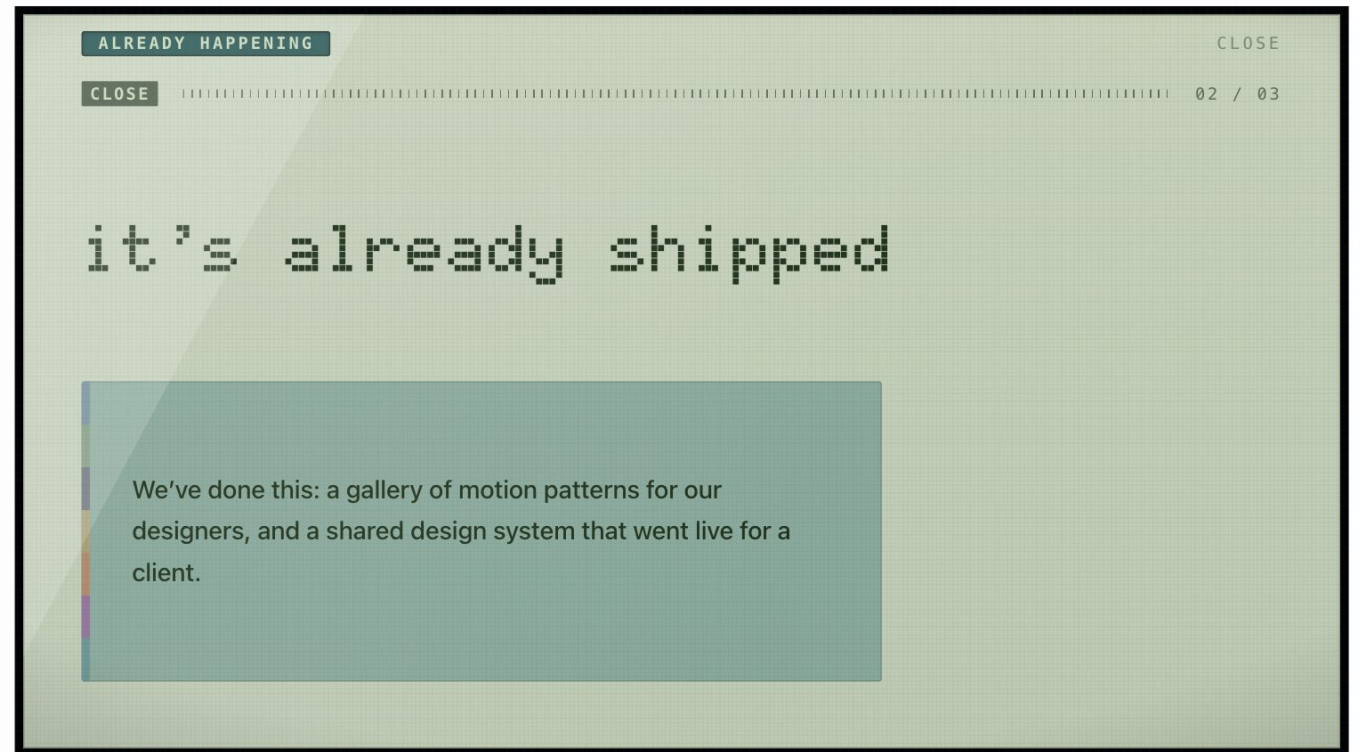
Real technical limits. No shared words for motion. Saying clearly what you want. No agreed limit for how much motion a page can afford.

- **Most of what you just watched was already a combination**
 - Real page elements where you need text, clicks, accessibility
 - The graphics card where you need pixels
 - An alternative for anyone who can't use the GPU layer
- **A performance strategy, not just an aesthetic one**
 - Each part on the cheapest surface that can carry it
- **Don't be a purist**
 - Purism forces one surface to do work it's bad at
- **For a brief**
 - Not "which technique is this project"
 - *Which technique carries which part?*



Most of what you just saw combined several of these. The useful question isn't which technique a project uses. It's which one does which part.

- **We've already been doing this, and it shipped**
- **Design-token and component system**
 - Shipped globally for a client
 - Isolate, then combine — the logic transfers to motion, per technology
- **Animotion**
 - A Storybook-style gallery of motion patterns for our designers
 - Easing, duration, Bézier curves → nav menus, page transitions
 - Primitives built up like tokens: atom → molecule → organism
 - **CSS, SVG and JavaScript only**
- **A progress report, not a proposal**
 - *The gap is opportunity, not skill*



We've done this: a gallery of motion patterns for our designers, and a shared design system that went live for a client.

- **Back to the question**

- *Could you describe this well enough for someone to build it?*
- You've now watched things you couldn't have described

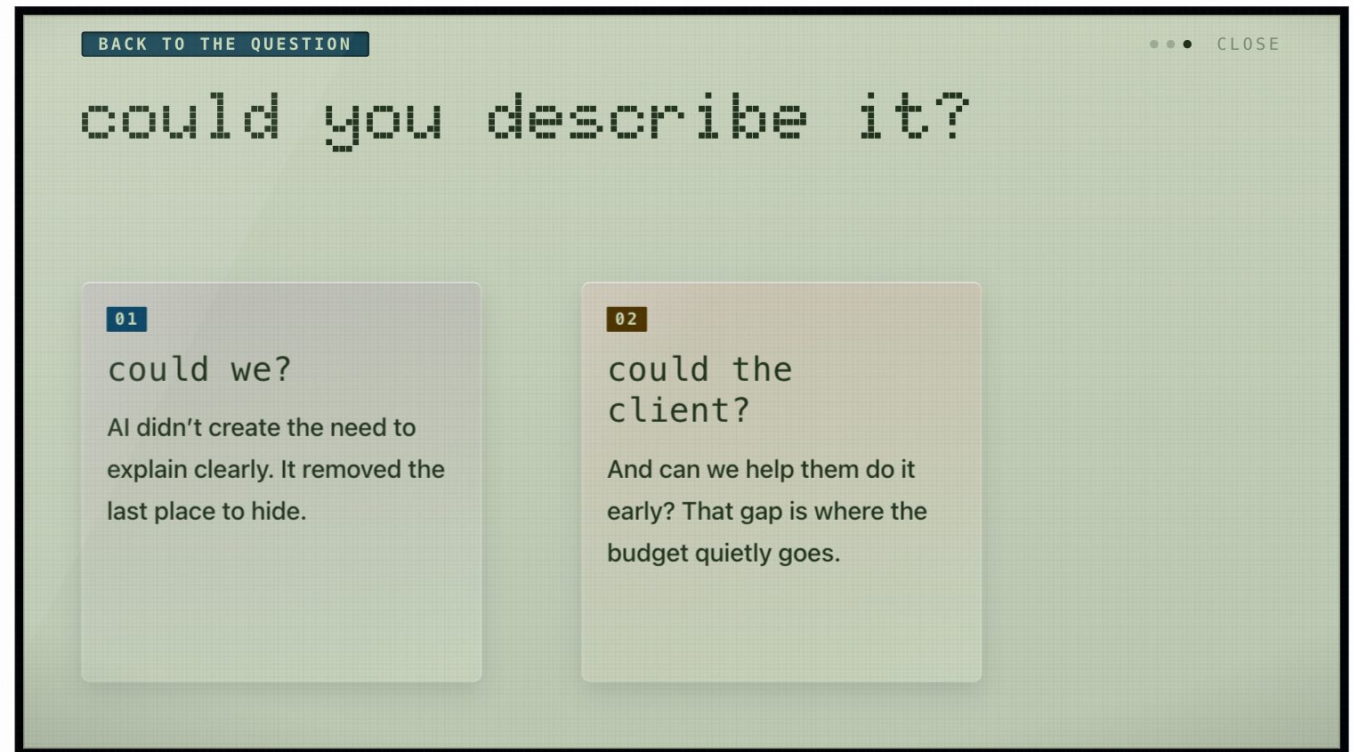
- ***AI didn't create the need to articulate. It removed the last place it could hide.***

- **The craft question**

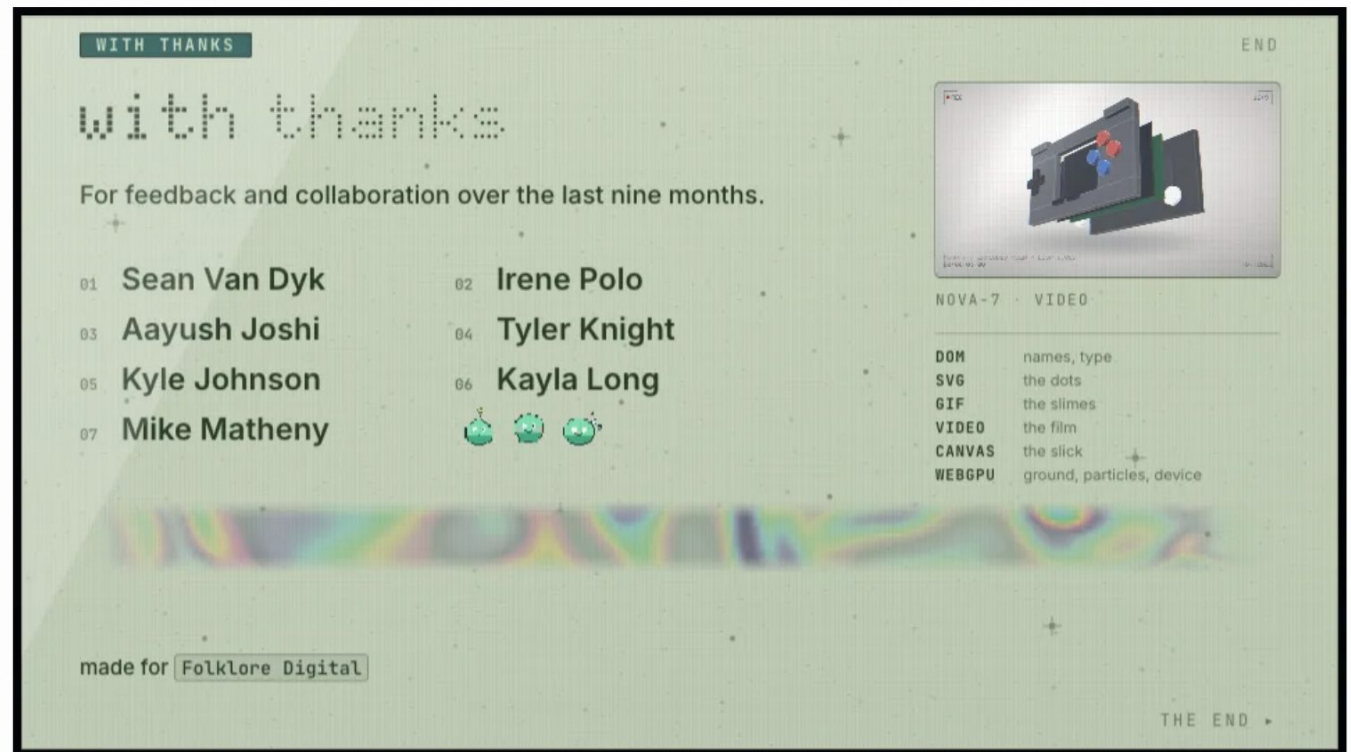
- Can **we** describe it?

- **The client question — the bigger one**

- Can a client who isn't technical or designerly describe what **they** want?
- Can we facilitate that conversation early enough not to fly blind?
- The gap between ideation and a living experience is where **budget quietly goes**



- **Everything you just saw, at once**
 - The credits are real page elements, SVG, GIFs, video and a 2D canvas composited over a WebGPU device
- **The whole talk was the composite**
- **With thanks**
 - Sean Van Dyk
 - Irene Polo
 - Aayush Joshi
 - Tyler Knight
 - Kyle Johnson
 - Kayla Long
 - Mike Matheny
- **For feedback and collaboration over the last nine months**



Live demo · composite/finale

- **Thank you**
 - Made for Folklore Digital
- **The paper**
 - Full argument and every source
 - Slide numbers match its bibliography
- **The site**
 - Every example runs there



Every source is in the paper, and every example runs on the site. Scan to open it.